Chapter 17, Problem 56P

Problem

The Fourier series trigonometric representation of a periodic function is

$$f(t) = 10 + \sum_{n=1}^{\infty}\left(\frac{1}{n^2+1}\cos n\pi t + \frac{n}{n^2+1}\sin n\pi t\right)$$

Find the exponential Fourier series representation of *f*(*t*).

Step-by-step solution

Step 1 of 1

$$f(t) = a_o + \sum_{n=1}^{\infty} a_n \cos n\omega_o t + b_n \sin n\omega_o t$$

$$C_n = \frac{a_n - jb_n}{2} = \frac{1-jn}{2\left(1+n^2\right)}$$

$$C_{-n} = \frac{1+jn}{2\left(1+n^2\right)} \qquad C_o = 10$$

$$f(t) = 10 + \sum_{n=1}^{\infty} C_n e^{jn\omega_o t} + C_{-n} e^{jn\omega_o t}$$

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_o t}$$

$$\boxed{f(t) = 10 + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1-jn}{2\left(1+n^2\right)} e^{jn\pi t}}$$